

Towards Reliable Markers and Personalized Interventions for Social Anxiety Using Mobile and Wearable Sensors

Nilesh Kumar Sahu
Indian Institute of Science Education
and Research Bhopal
Bhopal, Madhya Pradesh, India
nilesh21@iiserb.ac.in

Snehil Gupta
All India Institute of Medical Sciences
Bhopal
Bhopal, Madhya Pradesh, India
snehil.psy@aiimsbhopal.edu.in

Haroon R. Lone
Indian Institute of Science Education
and Research Bhopal
Bhopal, Madhya Pradesh, India
haroon@iiserb.ac.in

Abstract

Anxiety disorders are common among young adults, yet early detection and real-time intervention remain limited in practice. This research leverages mobile and wearable sensing to develop scalable anxiety detection and adaptive intervention systems. It identifies key markers of anxiety using physiological and audiovisual data and models user context using mobile sensor data. An adaptive intervention module tailors interventions to maximize user acceptability. This work contributes toward building intelligent, real-world, Just-In-Time Adaptive Intervention (JITAI) systems that support context-aware, personalized mental health care for social anxiety in everyday settings.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing; Ubiquitous and mobile devices; Smartphones; Mobile devices; Empirical studies in ubiquitous and mobile computing; User studies; Field studies; Laboratory experiments; Empirical studies in HCI**; • **Applied computing** → **Health informatics.**

Keywords

mHealth, Mobile Sensing, Passive Sensing, Wearable Devices, User Context Modeling, Anxiety Detection, Intervention, JITAI

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1 Introduction

Social Anxiety Disorder (SAD) is a common mental health condition affecting approximately 10–13% of the population [7, 17, 19]. It is characterized by an intense fear of being negatively evaluated in social situations, such as meeting new people or presenting in front of a group [1]. As a result, individuals with SAD often avoid social interactions or endure them with significant distress [25]. Individuals with SAD exhibit physical and behavioral symptoms, where physical symptoms include increased heart rate, sweating,

and shortness of breath, and behavioral symptoms include shyness, limited or no eye contact, and the display of negative emotions.

Due to the absence of a reliable marker for SAD, clinician judgment and self-reported scales are considered the gold standard for diagnosis. However, these gold standard methods are prone to various biases [18, 25]. Clinician judgments are subject to human interpretation and may be influenced by personal biases. Similarly, self-reports are vulnerable to recall bias, as individuals must accurately remember and evaluate their past experiences.

The ubiquity of mobile sensing technologies has enabled the unobtrusive capture of individual behavioral and physiological characteristics associated with mental health disorders [27]. Mobile sensing offers several advantages—being lightweight, suitable for use in everyday environments, and minimally intrusive, making it a promising approach for identifying reliable markers of anxiety disorders and delivering timely interventions. Today’s smartphones, equipped with a range of built-in sensors, can serve as objective tools for detecting anxiety symptoms in a manner that is neither burdensome nor invasive.

Such systems also support remote assessment, helping bridge the mental health treatment gap by identifying individuals needing prevention, evaluation, or therapeutic support. These services can be delivered either in person or via digital health (eHealth) platforms. Moreover, interventions can be administered directly through smartphones using specialized mHealth (mobile health) applications. The ultimate goal of mHealth is to integrate accurate sensing with effective interventions, thereby improving the quality of life for individuals affected by mental health conditions.

My PhD research aims to advance the field of mHealth in the context of anxiety disorders. Specifically, it focuses on two primary objectives: first, to identify reliable markers of anxiety using wearable and mobile sensing technologies, and second, to develop and deliver effective interventions via smartphones. The following subsection outlines the central research question that guides my doctoral work.

1.1 Problem Statement

1. Previous research on the physiological markers of anxiety disorders has primarily focused on heart rate variability (HRV) features [2, 4, 10, 16]. However, findings across studies have been inconsistent. Notably, Held et al. [6] argue that these mixed results are not solely due to the nature of the anxiety-inducing tasks but are also influenced by variations in participant characteristics. These individual differences raise important concerns about the generalizability of existing findings to broader populations. *My research is the first to examine physiological markers of*



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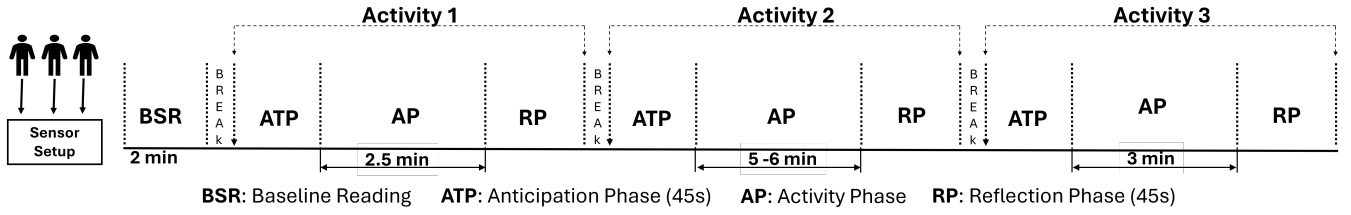


Figure 1: Study Flow: Activity 1, Activity 2, and Activity 3 correspond to Speech, Group Discussion, and Interview, respectively.

anxiety disorders within populations from the Global South and behavioral markers of anxiety disorders, addressing a significant gap in the current literature.

2. Previous research on mobile notifications has extensively examined optimal delivery times and conditions using passively collected data from mobile sensors and user activity patterns, such as location [9, 11, 12, 15, 26], physical activity [9, 11, 14, 15], and phone interaction behavior [9, 11]. However, these approaches often fall short when applied to the timing of behavioral intervention notifications [9, 11]. Unlike routine notifications, behavioral interventions demand a higher level of user receptivity. Even highly motivated users are likely to dismiss such prompts if they are delivered during moments of emotional unavailability or when the content does not align with the user's current state. *To address this limitation, my research leverages reinforcement learning to model user context and develop an adaptive intervention system that intelligently personalizes the type of behavioral prompts to maximize user engagement and acceptance.*
3. Building on the findings of the previous problem statements, this research aims to develop a JITAI [9, 11, 13] system for individuals with anxiety disorders. This system will utilize data from wearable sensors (e.g., smart watches) and mobile sensors to predict the onset and severity of anxiety symptoms in real time. Based on the predicted emotional state, current user context, and historical patterns of intervention acceptance (i.e., intervention likability), the adaptive intervention module will intelligently select and deliver the most appropriate intervention at the optimal moment. *This personalized approach aims to improve the intervention's effectiveness and overall user engagement.*

Collectively, this research will make three key contributions toward the development of intelligent, context-aware mental health support systems for anxiety disorders: (a) Demonstrate the feasibility of leveraging physiological signals (ECG¹, EDA²), audiovisual data, and mobile sensor data to identify reliable markers of anxiety—particularly among underrepresented populations from the Global South; (b) Establish the effectiveness of reinforcement learning-based adaptive intervention delivery, grounded in real-time contextual data and user feedback, to improve intervention receptivity in real-world settings; and (c) Lay the groundwork for a JITAI framework that integrates wearable and mobile sensing technologies to enable timely, personalized, and scalable mental health interventions.

¹ECG: Electrocardiogram

²EDA: Electrodermal activity

2 Methodology

To address my research questions, I have designed and conducted two studies so far, one in a controlled setting and the other in a real-world (wild) setting. I will describe both studies in detail below.

2.1 Study 1

Study 1 focuses on identifying reliable markers of SAD [20, 22]. The study involved three anxiety-inducing activities: a speech task, a group discussion (GD), and a mock interview. The speech task was based on the widely used Trier Social Stress Test (TSST), a standard protocol for eliciting physiological responses to stress and anxiety. The group discussion was designed to provoke anxiety related to fear of negative evaluation, perceived lack of control, heightened self-awareness, and social comparison. The mock interview task aimed to trigger anxiety through fear of scrutiny, performance evaluation, making mistakes, and potential rejection or negative consequences. Although all three activities are known to induce anxiety, each represents a unique social context that shapes participants' experiences differently.

This study adopted a novel structure where each activity was divided into three distinct phases: anticipation phase (ATP), activity phase (AP), and reflection phase (RP). During the ATP, participants anticipated the upcoming task, allowing us to capture their pre-task emotional and physiological states. In the AP, they actively engaged in the activity, while the RP involved reflecting on their performance, providing insight into post-task affective responses. This temporal segmentation allowed for a detailed investigation into how anxiety evolves over time across different social scenarios. The tasks were administered in a fixed sequence: first, the speech task, followed by the GD, and finally, the mock interview (see Figure 1).

Data Collection: The multimodal data were collected from 111 participants, including self-reported surveys, physiological signals, and audio-visual recordings. (i) *Physiological Data:* ECG, PPG, and EDA signals were recorded using Shimmer wearable sensors (ECG³ and GSR kits⁴, see Figure 2a and 2b) at 1024 Hz. (ii) *Audio-Visual Data:* Audio and video were recorded using three low-cost smartphones (one per participant, see Figure 2c), with audio sampled at 44 kHz and video at 27 FPS. (iii) *Self-Reported Measures:* Participants completed the SPIN questionnaire to assess SAD severity. Additionally, custom-validated state anxiety questionnaires were administered at baseline, during each activity phase (ATP, AP, and RP), and post-study to capture momentary anxiety levels.

³<https://www.shimmersensing.com/product/shimmer3-ecg-unit-2/>

⁴<https://www.shimmersensing.com/product/shimmer3-gsr-unit/>

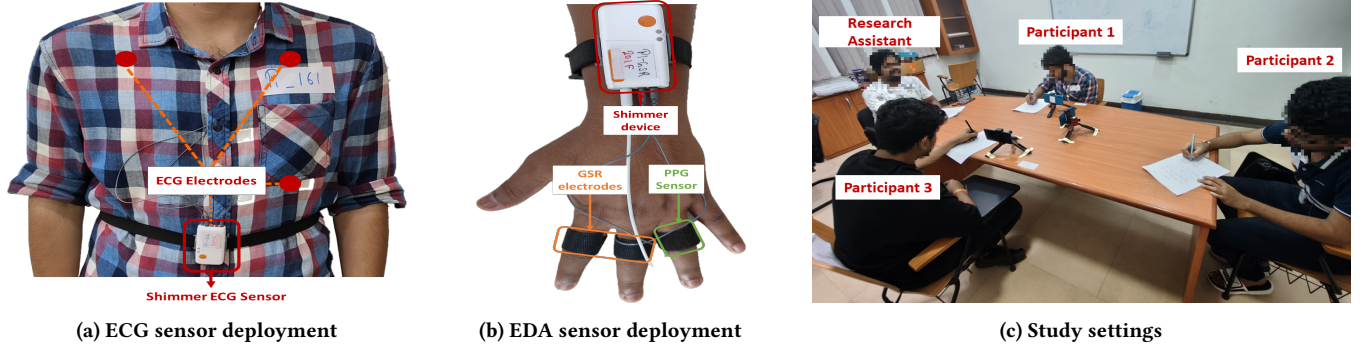


Figure 2: Sensor deployments and study settings

2.2 Study 2

Study 2 focused on designing a contextually adaptive intervention module for participants. For this, a custom Android application (LogMe) was used to collect passive data (e.g., mobile sensor) and active data (e.g., self-reported survey) and deliver behavioral interventions to promote student mental health and well-being (see Figure 3). The app delivered macro-level behavioral interventions 14 times daily during the last five minutes of each hour between 7:00 AM and 8:00 PM. Before intervention delivery, the application collected users' mood ratings using the Photographic Affect Meter (PAM) and contextual and social information (see Figure 3). The interventions were based on established research [3, 28] and validated by a clinical psychiatrist to ensure their relevance and effectiveness.

Adaptive intervention module: One well-known framework for intervention delivery is the JITAI [9, 13], which consists of two core components [11]: a just-in-time detection module and an adaptive intervention module. In this work, the focus was on the adaptive intervention module to evaluate user acceptability and the feasibility of real-time intervention delivery in naturalistic settings before integrating it into a complete JITAI system.

In this study, a participant's affirmative response to an intervention prompt was assumed as willingness and ability to engage with the suggested activity (see Figure 3d). The participant's response to each intervention prompt acted as a reward. This reward-based response framework allowed us to simultaneously capture the acceptability and feasibility of each intervention, providing valuable insights into how users evaluate interventions in real-time, real-world scenarios. Given its proven effectiveness in health promotion and behavior change settings [8], the Thompson Sampling algorithm was used to develop the adaptive intervention module. The algorithm used for building adaptive intervention is shown in Algorithm 1.

Data Collection: The data collection was conducted for two weeks with 77 participants. The application collected a rich dataset comprising passive mobile sensor data and active self-reported inputs. The mobile sensor data included readings from the accelerometer, gravity sensor, gyroscope, rotation vector, magnetometer, ambient light sensor, pedometer, proximity sensor, and GPS-based location. Device and usage data encompassed screen brightness, phone usage

Algorithm 1 Personalized Intervention Selection via Thompson Sampling (TS). Here α and β are the parameters of TS.

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1: Inputs:
    $\mathcal{A}$ : Set of activities
    $\mathcal{S}$ : Set of social contexts
    $\mathcal{I}_a$ : Set of candidate interventions for each activity  $a \in \mathcal{A}$ 
   For each activity  $a \in \mathcal{A}$  and intervention  $i \in \mathcal{I}_a$ , maintain:
      $\alpha_{a,i}$  and  $\beta_{a,i}$  (initialized using prior estimates)
2: for each interaction do
3:   Detect current activity  $a \in \mathcal{A}$ 
4:   Detect current social context  $s \in \mathcal{S}$ 
5:   for each intervention  $i \in \mathcal{I}_a$  do
6:     Sample  $\theta_{a,i} \sim \text{Beta}(\alpha_{a,i}, \beta_{a,i})$ 
7:   end for
8:   Select intervention  $i^* = \arg \max_{i \in \mathcal{I}_a} \theta_{a,i}$ 
9:   Deliver intervention  $i^*$ 
10:  Observe reward  $r \in \{0, 0.5, 1\}$  based on feedback
11:  Update Beta parameters:
12:     $\alpha_{a,i^*} \leftarrow \alpha_{a,i^*} + r$ 
13:     $\beta_{a,i^*} \leftarrow \beta_{a,i^*} + (1 - r)$ 
14: end for

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patterns, device mode, screen interactions, Wi-Fi connectivity, battery status, network state, system announcements, and interaction events. The active self-reports included mood ratings, contextual information such as the user's current activity, social context, categorized as being alone, with someone, or engaged in conversation, and intervention feedback.

3 Evaluation

3.1 Study 1

3.1.1 Analysis: Physiological and audiovisual data were analyzed separately. For physiological signals, standardized processing techniques were applied to ECG and EDA data, including filtering, peak detection, and other preprocessing steps [20]. From the ECG, HRV features were extracted across the time-domain, frequency-domain, and nonlinear dimensions. EDA signals were decomposed into phasic and tonic components, and time-domain features were derived accordingly. Inferential statistical analysis was conducted on the

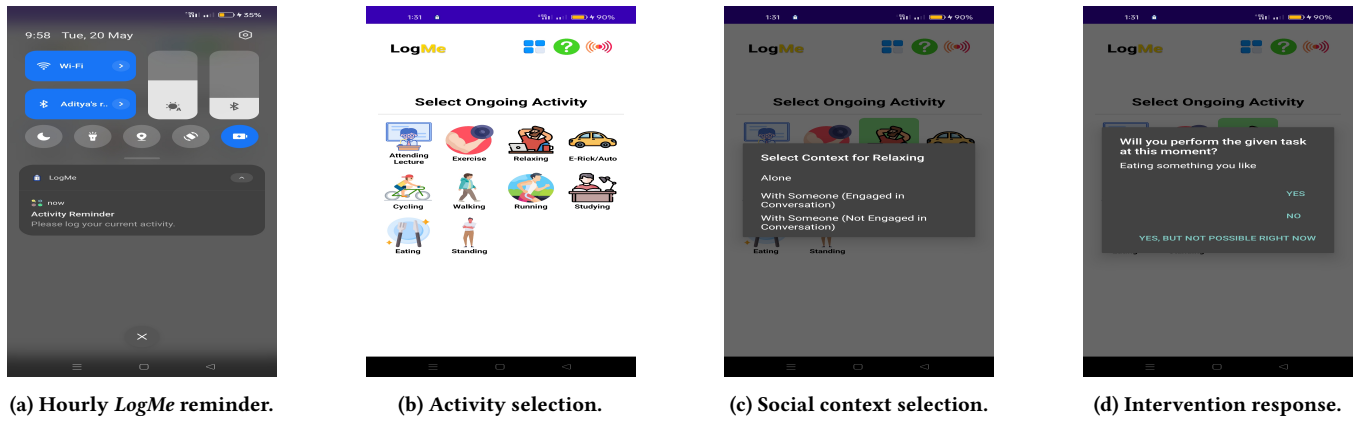


Figure 3: *LogMe* application design.

HRV features to identify potential physiological markers of anxiety, with comparisons made between individuals with SAD and those without (non-SAD). Machine learning (ML) models were also employed to classify participants based on their HRV profiles.

In the case of audio, both acoustic and linguistic features were extracted [24]. Acoustic features were computed using the Librosa Python library, while linguistic features were derived using LIWC software. In video data, facial action units, head pose, body posture, and eye gaze features were extracted using the openFace library [5, 21, 23]. Further, these features were used to build classification models identifying discriminative markers between SAD and non-SAD participants.

3.1.2 Findings: Statistical analysis indicated that ECG-derived HRV features are reliable physiological indicators of anxiety, while EDA features did not exhibit statistically significant group differences [20, 22]. To evaluate the performance of the classification models, 5-fold cross-validation was used across all modalities. The best results obtained were: (i) **HRV Features:** Accuracy = 71%, AUC = 0.68, (ii) **Audio Features:** Accuracy = 85.71%, AUC = 0.85, (iii) **Video Features:** Accuracy = 74%, F1-score = 0.70, (iv) **Eye Gaze Features:** Accuracy = 100%, F1-score = 1.00. *These findings highlight eye gaze features as particularly promising for capturing subtle behavioral indicators, achieving good classification performance when paired with deep learning techniques such as autoencoders.*

3.2 Study 2

3.2.1 Analysis: Features such as app category usage, intervention timing (day and hour), battery level, screen interactions, and location were extracted from the mobile sensor data. Inferential statistical methods were used to analyze self-reported data, identifying contextual factors influencing intervention acceptance and receptivity. Based on these insights, ML and DL methods were applied to automate intervention delivery. The objective was to optimize the timing and contextual relevance of prompts, thereby increasing the likelihood of user engagement.

3.2.2 Findings: Several contextual features—time of day, day of the week, battery level and status, screen status, lock/unlock events, physical activity, app usage category, call state and location—emerged

as significant predictors of intervention acceptability. These features were used to train ML and DL models. The highest accuracy of 75.8%, an F1-score of 0.856, and a Matthews Correlation Coefficient (MCC) of 0.174 was achieved to find the slot at which intervention if delivered, will be accepted by the user.

3.3 Future Work

- To improve the classification accuracy achieved in Study 1, future work will focus on multimodal fusion of physiological, audio, and video signals. Planned experiments will evaluate whether combining these modalities can yield more reliable and generalizable models for anxiety detection.
- Based on the insights from Study 2, the next phase will involve refining the context-aware intervention delivery model. Further, building a fully functional JITAI system, integrating real-time sensing, adaptive delivery, and user modeling. This system will be deployed in a longitudinal field study to assess its real-world effectiveness in supporting individuals with anxiety.

4 Conclusion

Mobile sensing and wearable technologies offer promising pathways for scalable, context-aware mental health support. This work presents the design and evaluation of approaches that leverage physiological signals, audiovisual features, and mobile sensor data for anxiety detection and personalized intervention delivery. Study 1 highlights the potential of ECG-derived HRV and eye gaze features as reliable markers for SAD. Study 2 introduces a contextually adaptive mobile intervention system that utilizes real-time behavioral and sensor data to optimize intervention timing. Based on insights from both studies, the JITAI system will be built to predict the onset of SAD and enable timely, personalized, and acceptable mental health support. Overall, this research contributes to the development of intelligent, low-burden mental health tools that can empower both users and clinicians by facilitating real-time detection and adaptive intervention in everyday settings.

References

- [1] Arya Adyasha, Anushka Sanjay Shelke, Nilesh Kumar Sahu, and Haroon R Lone. 2025. The Impact of Acute Social Stress on Working Memory Updating in Social

- Anxiety Disorder. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, Vol. 47.
- [2] Gail A Alvares, Daniel S Quintana, Andrew H Kemp, Anita Van Zwieten, Bernard W Balleine, Ian B Hickie, and Adam J Guastella. 2013. Reduced heart rate variability in social anxiety disorder: associations with gender and symptom severity. *PLoS one* 8, 7 (2013), e70468.
 - [3] Phoebe Barnett, Laura-Louise Arundell, Rob Saunders, Hannah Matthews, and Stephen Pilling. 2021. The efficacy of psychological interventions for the prevention and treatment of mental health disorders in university students: A systematic review and meta-analysis. *Journal of affective disorders* 280 (2021), 381–406.
 - [4] Paul Grossman, Frank H Wilhelm, Ichiro Kawachi, and David Sparrow. 2001. Gender differences in psychophysiological responses to speech stress among older social phobics: congruence and incongruence between self-evaluative and cardiovascular reactions. *Psychosomatic Medicine* 63, 5 (2001), 765–777.
 - [5] Nandigramam Sai Harshit, Nilesh Kumar Sahu, and Haroon R Lone. 2024. Eyes Speak Louder: Harnessing Deep Features From Low-Cost Camera Video for Anxiety Detection. In *Proceedings of the Workshop on Body-Centric Computing Systems*. 23–28.
 - [6] Judith Held, Andreea Visla, Christine Wolfer, Nadine Messerli-Bürgy, and Christoph Flückiger. 2021. Heart rate variability change during a stressful cognitive task in individuals with anxiety and control participants. *BMC psychology* 9, 1 (2021), 1–8.
 - [7] Ronald C Kessler, Maria Petukhova, Nancy A Sampson, Alan M Zaslavsky, and Hans-Ulrich Wittchen. 2012. Twelve-month and lifetime prevalence and lifetime morbid risk of anxiety and mood disorders in the United States. *International journal of methods in psychiatric research* 21, 3 (2012), 169–184.
 - [8] Harsh Kumar, Tong Li, Jiakai Shi, Ilya Musabirov, Rachel Kornfield, Jonah Meyerherhoff, Ananya Bhattacharjee, Chris Karr, Theresa Nguyen, David Mohr, et al. 2024. Using adaptive bandit experiments to increase and investigate engagement in mental health. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 22906–22912.
 - [9] Florian Künzler, Varun Mishra, Jan-Niklas Kramer, David Kotz, Elgar Fleisch, and Tobias Kowatsch. 2019. Exploring the state-of-receptivity for mHealth interventions. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 4 (2019), 1–27.
 - [10] Iris B Mauss, Frank H Wilhelm, and James J Gross. 2003. Autonomic recovery and habituation in social anxiety. *Psychophysiology* 40, 4 (2003), 648–653.
 - [11] Varun Mishra, Florian Künzler, Jan-Niklas Kramer, Elgar Fleisch, Tobias Kowatsch, and David Kotz. 2021. Detecting receptivity for mHealth interventions in the natural environment. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 5, 2 (2021), 1–24.
 - [12] Leanne G Morrison, Charlie Hargood, Veljko Pejovic, Adam WA Geraghty, Scott Lloyd, Natalie Goodman, Danius T Michaelides, Anna Weston, Mirco Musolesi, Mark J Weal, et al. 2017. The effect of timing and frequency of push notifications on usage of a smartphone-based stress management intervention: an exploratory trial. *PLoS one* 12, 1 (2017), e0169162.
 - [13] Inbal Nahum-Shani, Shawna N Smith, Bonnie J Spring, Linda M Collins, Katie Witkiewitz, Ambuj Tewari, and Susan A Murphy. 2018. Just-in-time adaptive interventions (JITAs) in mobile health: key components and design principles for ongoing health behavior support. *Annals of Behavioral Medicine* 52, 6 (2018), 446–462.
 - [14] Mikio Obuchi, Wataru Sasaki, Tadashi Okoshi, Jin Nakazawa, and Hideyuki Tokuda. 2016. Investigating interruptibility at activity breakpoints using smartphone activity recognition API. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct*. 1602–1607.
 - [15] Martin Pielot, Bruno Cardoso, Kleomenis Katevas, Joan Serra, Aleksandar Matic, and Nuria Oliver. 2017. Beyond interruptibility: Predicting opportune moments to engage mobile phone users. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 3 (2017), 1–25.
 - [16] Andre Pittig, Joanna J Arch, Chi WR Lam, and Michelle G Craske. 2013. Heart rate and heart rate variability in panic, social anxiety, obsessive-compulsive, and generalized anxiety disorders at baseline and in response to relaxation and hyperventilation. *International journal of psychophysiology* 87, 1 (2013), 19–27.
 - [17] Parnava Das Preeti. 2019. Prevalence of social anxiety disorder and its determinants among undergraduate medical students of East Delhi. *Int J Community Med Public Health* 6, 3 (2019), 1335.
 - [18] Haroon Rashid, Sanjana Mendu, Katharine E Daniel, Miranda L Beltzer, Bethany A Teachman, Mehdi Boukhechba, and Laura E Barnes. 2020. Predicting subjective measures of social anxiety from sparsely collected mobile sensor data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 3 (2020), 1–24.
 - [19] Ayelet Meron Ruscio, Timothy A Brown, Wai Tat Chiu, Jitender Sareen, Murray B Stein, and Ronald C Kessler. 2008. Social fears and social phobia in the USA: results from the National Comorbidity Survey Replication. *Psychological medicine* 38, 1 (2008), 15–28.
 - [20] Nilesh Kumar Sahu, Snehil Gupta, and Haroon Lone. 2024. Wearable Technology Insights: Unveiling Physiological Responses During Three Different Socially Anxious Activities. *ACM Journal on Computing and Sustainable Societies* 2, 2 (2024), 1–23.
 - [21] Nilesh Kumar Sahu, Snehil Gupta, and Haroon R Lone. 2025. AnxietyFaceTrack: A Smartphone-Based Non-Intrusive Approach for Detecting Social Anxiety Using Facial Features. *arXiv preprint arXiv:2502.16106* (2025).
 - [22] Nilesh Kumar Sahu, Snehil Gupta, and Haroon R Lone. 2025. Exploring Heart Rate Variability and Heart Rate Dynamics Using Wearables Before, During, and After Speech Activity: Insights from a Controlled Study in a Low-Middle-Income Country. In *Proceedings of the ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies*. 362–378.
 - [23] Nilesh Kumar Sahu, Nandigramam Sai Harshit, Rishabh Uikey, and Haroon R Lone. 2024. Beyond Questionnaires: Video Analysis for Social Anxiety Detection. *arXiv preprint arXiv:2501.05461* (2024).
 - [24] Nilesh Kumar Sahu, Manjeet Yadav, and Haroon R Lone. 2024. Unveiling Social Anxiety: Analyzing Acoustic and Linguistic Traits in Impromptu Speech within a Controlled Study. *ACM Journal on Computing and Sustainable Societies* 2, 2 (2024), 1–19.
 - [25] Asif Salekin, Jeremy W Eberle, Jeffrey J Glenn, Bethany A Teachman, and John A Stankovic. 2018. A weakly supervised learning framework for detecting social anxiety and depression. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 2, 2 (2018), 1–26.
 - [26] Hillol Sarker, Moushumi Sharmin, Amin Ahsan Ali, Md Mahbubur Rahman, Rummana Bari, Syed Monowar Hossain, and Santosh Kumar. 2014. Assessing the availability of users to engage in just-in-time intervention in the natural environment. In *Proceedings of the 2014 ACM international joint conference on pervasive and ubiquitous computing*. 909–920.
 - [27] Zhiyuan Wang, Maria A Larrazabal, Mark Rucker, Emma R Toner, Katharine E Daniel, Shashwat Kumar, Mehdi Boukhechba, Bethany A Teachman, and Laura E Barnes. 2023. Detecting social contexts from mobile sensing indicators in virtual interactions with socially anxious individuals. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 7, 3 (2023), 1–26.
 - [28] Joanne Worsley, Andy Pennington, and Rhiannon Corcoran. 2020. What interventions improve college and university students' mental health and wellbeing? A review of review-level evidence. *National Grey Literature Collection* (2020), 1–54.